|INCH-POUND|
MIL-P-85548B(AS)
30 November 1990
SUPERSEDING
MIL-P-85548A(AS)
26 March 1986

MILITARY SPECIFICATION

PROPELLANT, COMPOSITE, CAST, N-60

This specification is approved for use by the the Naval Air Systems Command, Department of the Navy, and is available for use by the Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the manufacture and acceptance of one type of cast composite propellant, N-60, referred to herein as the propellant.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-192	Ammonium Perchlorate
DOD-D-23443	Di-(2-Ethylhexyl) Adipate, Technical
MIL-I-85370	Iron Oxide,(Monohydrate) Yellow
MIL-A-85495	Antioxidant, T-Butylphenol Type 2, 2- Methylenebis (4-Methl-6-T-Butylphenol)
MIL-R-85545	Rocket, GTR-18A

STANDARDS

MILITARY

MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
DOD-STD-2101	Classification of Characteristics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1370
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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, Pa 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

1335AS280	Rocket, Loaded GTR-18A
1335AS286	Zinc Powder
1335AS287	Dicyclohexylmethane 4, 4'-diisocyanate
1335AS288	Cure Catalyst; Triphenyl Bismuth

(Unless otherwise indicated, copies of Naval Air Systems Command drawings are available from the Naval Air Technical Services Facility (NATSF) (Code 3121), 70C Robbins Avenue, Philadelphia, Pa 19111-5096.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

WS 20700	Polybutadiene, Liquid, Hydroxyl Terminatec

(Copies are available from the Naval Ordnance Station, Technical Documents (Code 802), Louisville, KY 40214-50001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

PUBLICATIONS

CHEMICAL PROPULSION INFORMATION AGENCY

CPIA, Publication No. 21	The Solid Propellant Mechanical Behavior: Method for Determining Uniaxial Tensile Test at Constant Strain Rate

(Applications for copies should be addressed to the Chemical Propulsion Information Agency, Applied Physics Laboratory, Johns Hopkins University, Johns Hopkins Road, Laurel, Maryland 20810.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 330	Average Particle Size of Refractory Metals and Compounds by Fisher Subsieve
ASTM D 792	Specific Gravity and Density of Plastics by Displacement

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103

(Non-Government standards and other publications are normally available for the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the test of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Composition. Composition of the propellant and requirements for each material shall be as specified in table I.

TABLE I. Composition.

Material 1/	Requirements	Percent by weight 3/
Hydroxyl terminated polybutadiene (HTPB)	WS 20700	[11.97] 2/
2, 2'-Methylenebis-(4-methyl-6 tertiary butylphenol) (DMMBP)	MIL-A-85495	
Dicyclohexyl methane 4, 4'-diisocyanate (H_{12}MDI)	1335AS287	
Di-(2-ethylhexyl)adipate (DOA)	DOD-D-23443	3.000
Triphenyl bismuth	1335AS288 4/	0.030
Ferric oxide (yellow) Fe_2O_3/H_2O)	MIL-I-85370	1.000
Zinc Powder	1335AS286 5/	40.000
Ammonium perchlorate (AP), Grade C bimodal blend (see table III)	MIL-A-192 6/	44.000

1/ For possible sources see 6.5.

2/ The total of all three materials is 11.97%. See 6.6 and 6.6.1 for calculations.

3/ See table II.

4/ See 6.7.

5/ See 3.3.

6/ See table III.

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TABLE II. Weighing accuracies.

Material	Tolerances (± %) 1/
HTPB & DMMBP mixture	
HTPB	2.5
DMMBP	2.5
Propellant mixture	
HTBP-DMMBP	0.5
H_{12}MDI	0.25
DOA	0.25
Triphenyl bismuth	1.5
Ferric oxide	0.25
Zinc powder	0.5
AP	0.5

1/ % of weight of material in the mixture

TABLE III. Ammonium perchlorate composition.

Component nominal size	Ratio in blend 1/	MIL-A-192 Grade	MIL-A-192 Class	Comments
Ground 6 ± 2 micron	61.37 - 75.00	C	N/A	See 4.6.1 and 4.6.3
200 micron	38.63 - 25.00	C	4	Round type

1/ May be varied within the range shown to adjust burning rates to the requirements of 3.4.3.

3.3 Gassing of zinc powder. The zinc powder shall not exhibit gassing when tested according to 4.6.2.

3.4 Chemical, mechanical, and ballistic properties. The cured propellant samples used for the determination of the characteristics listed below shall be stored at 77° ± 10° F in a condition where moisture absorption is minimized. Excursions from these conditions for periods up to 8 hours for handling, transportation and machining are permissible. The propellant shall not exceed a stored life of 2 weeks between the end of cure and the start of analysis.

3.4.1 Density (M101). The density of the cured propellant shall be 2.23 ± 0.03 grams per cubic centimeter (gm/cm^3) at 77° ± 3° F (see 4.7.1).

3.4.2 Mechanical properties. The mechanical properties of the cured propellant shall be as follows (see 4.7.2):

a. Tensile Strength: 100 pounds per square inch (psi) minimum
b. Elongation: Not less than 8 percent at maximum tensile load

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3.4.3 Strand burning rate. The burning rate, (r_b) in inches per second, shall be obtained as specified in 4.7.3 and shall fall within the band specified by the following equation:

$$r_b = AP^{0.4} \text{ inches/second}$$

where:

A = 0.030 ± 0.006 ((inches/second) $(psia)^{-0.4}$)
P = average actual pressure (psia) (see 4.7.3)

The equation is valid for pressures between 15 and 50 pounds per square inch absolute psia at 77° ± 3°F (see 4.7.3).

3.4.4 Heat of explosion (MIO2). The heat of explosion shall be 650 ± 35 calories per gram (cal/gm) (see 4.7.4).

3.5 Stability. During the time the tests specified herein are performed, the propellant shall not exhibit any physical, mechanical, or ballistic changes from start to completion of tests.

3.6 Identification and marking. Each propellant batch shall be assigned an identification designation, which shall include the date of mix. Each chemical used in the batch shall be recorded by its name, production lot number, and quality assurance department certification. The marking of any propellant samples, containerized propellant, or other items which contain propellant shall have the propellant's batch designation clearly shown.

3.7 Workmanship. The propellant shall be free from any contamination or foreign material that would render the propellant unsuitable for the propellant's intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), inspection conditions shall be as follows:

Temperature:	65° to 90° F
Humidity:	95 percent, maximum

4.4 First article inspection. First article inspection shall be performed by the contractor after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production (see 6.3).

4.4.1 Sampling. Unless otherwise specified in the contract (see 6.2), the first article sample shall consist of a block of propellant which has been vacuum cast and cured in accordance with note 3 of Drawing 1335AS280. The propellant sample shall be large enough for all first article samples required in table IV, but not larger than 700 cubic inches.

4.4.2 Inspection routine. The first article sample shall be subjected to the tests specified in Table IV.

4.4.3 Failure. Failure of the sample to meet any of the requirements shall be cause for first article rejection.

TABLE IV. Inspection routine.

Tests	Number of inspection units 1/		Acceptance criteria	
	First article	Quality conformance	Accept	Reject
Density	10	2	0	1
Mechanical properties	10		0	1
Strand burning rate	9		0	1
Heat of explosion	10	2	0	1

1/ See 6.9.2 for definition.

4.5 Quality conformance inspection. Quality conformance inspection shall be performed on each lot of propellant (see 6.3).

4.5.1 Sampling. Unless otherwise specified in the contract (see 6.2), the quality conformance sample shall consist of a block of propellant which has been vacuum cast and cured in accordance with note 3 of Drawing 1335AS280. The propellant sample shall be large enough for all quality conformance samples required in table IV, but not larger than 300 cubic inches.

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4.5.2 Lot. Unless otherwise specified in the contract, a lot shall consist of all the propellant manufactured in one batch from a single lot of each raw material.

4.5.3 Inspection routine. The inspection routine shall be as specified in table IV.

4.5.4 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2101. Requirements that are critical are identified by the symbol (C) and those that are major are identified by the symbol (M). The number following the classification symbol indicates the serial number of the requirement. Requirements not annotated with a classification code are classified minor.

4.6 Test methods for material.

4.6.1 Particle size of ammonium perchlorate. The 6 micron (nominal) ammonium perchlorate shall be tested for particle size in accordance with ASTM B 330.

4.6.2 Gassing of zinc powder. Each lot of zinc powder shall be tested for gassing as follows: Mix 75 grams of HTPB (containing 0.99 percent DMMBP), 8.8 grams of H_{12}MDI, 0.2 grams triphenyl bismuth, and 280 grams of zinc powder. Apply a vacuum and place in a 135° ± 5° F oven for 24 ± 4 hours. Remove sample from oven and section it. The zinc powder will have settled and the portion of the sample with the settled zinc will have a large number of small bubbles if the zinc causes gassing. If the zinc powder exhibits gassing, it may be degassed and retested.

4.6.3 Surface moisture requirement of ammonium perchlorate. The 6 micron (nominal) ammonium perchlorate shall not be greater than a weight of 0.030 percent when tested in accordance with MIL-A-192.

4.7 Test methods.

4.7.1 Density. The density shall be determined on test specimens in accordance with ASTM D 792, Method A-2. The test temperature shall be 77° ± 3° F. An optional method for determining the density of the immersion liquid (see 6.8) is to test a piece of glass of accurately known density in the same manner as the specimens.

$$\text{Density of immersion liquid} = \frac{(WA - WL)\,D}{WA}$$

where:

WA = weight of glass standard in air
WL = apparent weight of glass standard while immersed in liquid
D = density of glass standard

4.7.2 Mechanical properties test. Tensile strength and elongation at maximum tensile load shall be determined on Class B or Class C tensile test specimens in accordance with the procedures for preparation, storing, conditioning, and testing of tensile specimens given in section 4.3 of CPIA Publication No. 21, Class B- "Quality Control Testing" or Class C- "Development Screening Testing." These two methods shall give equivalent results for the material. The specimen

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shall be tested using a crosshead separation rate of 2 inches per minute and at a test temperature of 77° ± 2° F and 85 percent maximum relative humidity. An effective gauge length of 2.7 inches shall be used to calculate elongation.

4.7.3 Strand burning rate. Strand burning rate on cured specimens shall be determined according to MIL-STD-286, method T803.1. Strands shall be inhibited with polyester inhibitor (Koppers 6065-5 Polyester Resin catalyzed with 1-1/2 percent MEK peroxide (60 percent MEK peroxide in dimethylphthalate), or equivalent). A minimum of three strands shall be tested at each of three pressures: 15 + 5, - 0, 30 ± 5, and 45 ± 5 psia. Strands shall be burned at 77° ± 3° F. The starting pressure and the maximum pressure attained shall be recorded and averaged. The pressure gage for this test shall have an accuracy of 1 psia and the timer shall have an accuracy of 0.5 seconds.

4.7.4 Heat of explosion. This test shall be performed using an adiabatic calorimeter such as the Parr Instrument Company, Series 1200. A Parr No. 1101 or 1108 double valve oxygen bomb, or equal, shall be used. The calorimeter shall be standardized with benzoic acid using oxygen according to the manufacturer's instruction manual. The heat of explosion test shall be conducted in the same manner as the standardization with the following limitations and exceptions.

a. Water shall not be added to the bomb.

b. The bomb shall be purged three times with 15 to 20 atmospheres of helium or argon; then pressurize the bomb to 25 ± 2 atmospheres of helium or argon for testing.

NOTE: Impurities in the inert gas such as oxygen or water can cause systematic high results. Each new bottle should be tested for suitability for this test by comparison with gas of known purity.

4.7.4.1 Calculation.

$$\text{Heat of explosion, cal/gm} = \frac{(t_2 - t_1)\,W}{m}$$

where:

t_2 = temperature of calorimeter at end of test, degrees Celsius (°C).
t_1 = temperature of calorimeter before ignition, °C.
W = energy equivalent factor determined by standardization as above, calories per °C.
m = weight of sample, grams(s).

4.7.4.2 Specimens. A heat of explosion test specimen is defined as a single piece of solid propellant 2.0 ± 0.5 grams in weight, weighted to the nearest 0.0001 gram.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The propellant covered by this specification is intended for use in the GTR-18A rocket.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. If first article is necessary (see 3.1).

d. Inspection conditions if other than specified in 4.3.

e. Quality conformance sampling if other than specified in 4.5.1.

f. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DoD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjuction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requiremenr for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4 and 4.5	DI-NDTI-80566	Test plan	---
4.4 and 4.5	DI-T-2072	Report, test	---

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of a block of propellant as specified in 4.4.1. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government

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approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Possible sources of supply. Products that have met the requirements of this specification in past procurement actions are listed below. This information is to assist in procurement and is no way restrictive. It is given for the information of the acquring activity and is not to be construed as a waiver of any requirement of this specification nor as any limitation of additional supply.

Material	Possible Source
Hydroxy-Terminated Polybutadiene (R-45HT)	ARCO Chemical Co. Division Atlantic Richfield 5505 E. Carson, Suite 255 Lakewood, CA 90713
Dicyclohexylmethane 4, 4' -diisocyanate (Desmodur W)	Mobay Chemical Co. Penn Lincoln Parkway West Pittsburgh, PA 15205
Di-(2-Ethylhexyl) Adipate (DOA)	HATCO Chemical Division W.R. Grace and Co. King George Post Road Fords, NJ 08863
2, 2'-Methylenebis-(4-methyl-6-tertiary-butylphenol (CAO-5)	Sherwin Williams Co. Chemical Division Fords, NY 03863
Triphenyl Bismuth	Boulder Scientific Co. P.O. Box 5480 Mead, CA 80542
Ammonium Perchlorate (AP)	Kerr McGee Chemical Corp. Kerr McGee Blvd. Oklahoma City, OK 73102
Zinc Powder	LDL Technology, Inc. 137 Pennsylvania Avenue Paterson, NJ 07503
Ferric Oxide, Yellow ($Fe_2O_3H_2O$) (Yellow Iron Oxide)	Cities Service Co. 1440 N. Harbor Blvd. Suite 421 Fullerton, CA 92635

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6.6 Composition calculation. The DMMBP is added to the HTPB and allowed to dissolve before mixing. One part of DMMBP is added to 100 parts of HTPB and this mixture is used in the mix. Weight % of the HTPB & DMMBP mixture and $H_{12}MDI$ are calculated as follows (see 6.6.1 for sample calculation):

$$(\%\ \text{HTPB \& DMMBP mixture}) = \frac{11.97}{((NCO/OH)\ (Ex + 0.00588)/(1.01\ Ey)) + 1}$$

$$\%\ H_{12}MDI = 11.97 - (\%\ \text{HTPB \& DMMBP mixture})$$

where

HTPB & DMMBP is a 100 part HTPB to 1 part DMMBP mixture

Ex = OH equivalency/100 gm of HTPB as determined by laboratory for the lot of HTPB used

Ey = NCO equivalency/100 gm of $H_{12}MDI$ as determined by laboratory for the lot of $H_{12}MDI$ used

(NCO/OH) = Cure ratio. 0.90 to 1.10 will give near optimum tensile strength and hardness for the propellant. The cure ratio may be varied, within 0.90 to 1.10 if needed, so that physical properties are met.

6.6.1 Sample calculation. The percentages of H12MDI and HTPB & DMMBP mixture used in each particular mix are determined as follows:

a. The -NCO (isocyanate) and -OH (hydroxyl) equivalent weights are determined by laboratory analyses for the lots of $H_{12}MDI$ and HTPB to be used in the mix.

b. The following three equations are solved for percent IDPI and percent HTPB & DMMBP mixture.

(1) $$\frac{(\text{NCO equivalents/100 grams})\ (\%\ H_{12}MDI)}{(\text{OH equivalents/100 grams})\ (\%\ \text{HTPB}) + (0.588)\ (\%\ \text{DMMBP})} = 0.90 \text{ to } 1.10$$

(2) $(\%\ \text{HTPB \& DMMBP mixture}) = (1.01)\ (\%\ \text{HTPB})$

(3) $(\%\ H_{12}MDI) + (\%\ \text{HTPB \& DMMBP mixture}) = 11.97\%$

Example:

Given: $H_{12}MDI$ 0.759 NCO equivalents/100 grams
HTPB 0.081 OH equivalents/100 grams

Let

X = % HTPB & DMMBP mixture
X_1 = % HTPB
X_2 = % DMMBP
Y = % $H_{12}MDI$

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Then

$$X_1 + X_2 = X; \quad X = 1.01X_1; \quad \text{and } X + Y = 11.97$$

and

$$\frac{0.759\ (11.97 - X)}{0.081\ (X/1.01) + 0.588\ (X - X/1.01)} = 1.026$$

or

$$X = \frac{11.97}{\frac{(1.026)(0.081 + 0.00588)}{(1.01)\ (0.759)} + 1.0} = 10.723\%$$

therefore

$X_1 = 10.723/1.01 = 10.617\%$
$X_2 = 10.723 - 10.617 = .106\%$
$Y = (11.97 - X) = 1.247\%$

6.7 Solvent for triphenyl bismuth. Grade A dichloromethane, in accordance with MIL-D-6998, may be used as a solvent for the triphenyl bismuth. This solvent shall be removed (evaporated) from the composition during mixing at low pressure.

6.8 Immersion liquid. N-hexane has been found to be a satisfactory immersion liquid.

6.9 Definitions. For the purpose of this specificaton, the following definitions apply.

6.9.1 Batch. A batch is defined as that quantity of material which at one time has been subjected to one or more chemical or physical processes (or combination thereof) intended to produce a product having uniform characteristics.

6.9.2 Sample unit. A sample unit is a unit of propellant which is subjected to a first article or quality conformance test. The size of the unit depends on the test and is cited in the applicable test paragraph or in a document referenced in the test paragraph.

6.9.3 Cure ratio.

$$\text{Cure ratio} = \frac{(\text{NCO equivalents/100 grams})\ (\%\ H_{12}\text{MDI})}{(\text{OH equivalents/100 grams})\ (\%\ \text{HTPB}) + (0.588)\ (\%\ \text{DMMBP})}$$

6.10 Subject term (key word) listing.

Ammonium perchlorate
HTPB (Hydroxyl terminated polubutadiene)
Heat of explosion

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6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project 1370-N323)

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